SWEEP rule on the KD-Toric code

February 9, 2026

1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex v looks for a suggestion $\hat{v}$ on $\uparrow(v) \cap lk_2$ that matches σ_v^+. If there are more than one, then pick the one which has the minimal weight.

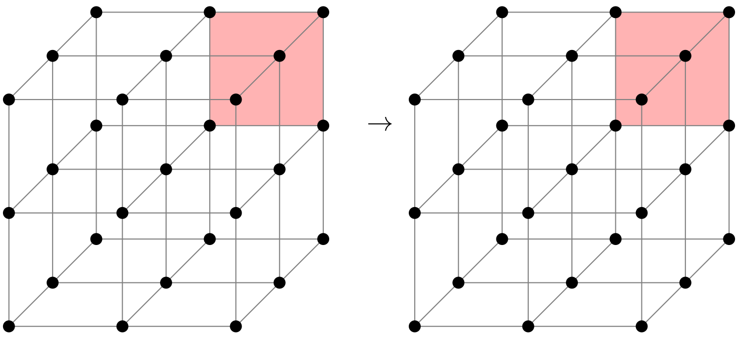

Figure 1: SWEEP rule

2 Span Expansion

Define the potentials functions $L_i : \mathbb{R}^d \to \mathbb{R}$ as follow[1]:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d+1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d+1 \end{cases}$$

The **Size** of a subset S is defined as:

$$\mathbf{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let e be an unsatisfied check at time t, and let H_e be the unsatisfied checks at time $t-1$. Then there is a set $H'_e \subset H_e$ such that:

1. For any $i < d+1$ there exists $e' \in H'_e$ such $L_i(e') \geq L_i(e)$.
2. There exists $e' \in H'_e$ such that $L_{d+1}(e') \leq L_{d+1}(e) + 1$.

Proof. Consider a vertex v which at time t adjoins to an unsatisfied edge e, let u, w be vertices which have a positive faces that adjoin to e.

[1]Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.

1. If $\hat{u}|_e \neq \emptyset$. Idea: At least two (tree in 4D) directions are taken by faces on u, then we track over a faces paths that is positive in the remain direction, we know that we always can go forward, otherwise we close a stabilizer what contradicts .. (Require refine, cause we can go backward yet don't expect to back to the starting point).

   Let $G \subset \mathcal{F}$ be a connected component of faces that contains b_u. We say that G is d far from u if

   $$\max_{f \in G} |f_x - u_x| \leq d$$

   Now, we will prove by induction on d that there is a face $f \in G$ with $f_x \geq v_x$ such one of its edge is not satisfied.

   (a) **Base.** The edge from v which point at the $\hat{z}$ direction must not be satisfied.

   (b) **Assumption.** Assume that for any G which is at most d far from u there is a face, with adjoins edge that is both has a x-coordinate that is greater than u_x and is not satisfied.

   (c) **Induction Step.** Notice that if by substitute a stabilizer from G we get G' that is at most d' far from u for some $d' < d$ then we done.

2. Else, if for any u it holds that $\hat{u}|_e = \emptyset$, then it must be that $e \in H_e$ because $\hat{v}$ can only decrease the syndrome on v. Moreover, it follows that $\hat{v} = \emptyset$ (otherwise, the suggestion would zero out the syndrome on $\uparrow(v) \cap lk_2$). Thus, $\sigma_v \not\subset \uparrow(v) \cap lk_2 \Rightarrow$ there is also an edge $e' \in H_e$ where $e' \in \downarrow(v)$. In that case, take $e \in H_e$ as the edge satisfies the first d equalities and e' for satisfying the $d+1$th inequality.

□

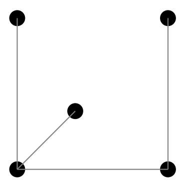

Figure 2: SWEEP rule